A MATHEMATICAL FIELD GUIDE · FIVE SYSTEMS · THREE TIMESCALES

The Learning Brain

as a system of equations

A single plasticity rule underlies the whole nervous system; the brain runs **five copies of it** with different teaching signals, coupled by activity but **cut off from each other's gradients**, gated **hour by hour** by the sleep–wake cycle, and rewired **decade by decade** by growth and pruning. This document assembles the complete formalism — every variable named — and grounds each piece in the cell types, densities, and dynamics that implement it.

THE UNIVERSAL SKELETON

$$\Delta w_{ij} = \eta \underbrace{\ell_{ij}}_{\text{local}} \underbrace{m_i}_{\text{teaching}}$$

Every rule below is this equation. The five systems differ only in what the *local factor* ℓ_{ij} and the *teaching factor* m_i are, and in which cells physically deliver m_i. Read the whole guide as variations on this line.

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00 · CONVENTIONS

Notation and the three clocks

Everything that follows uses one set of symbols and distinguishes three timescales on which the same synapse changes.

OPERATORS

$x^\top$	transpose of a vector or matrix
$a \odot b$	Hadamard (elementwise) product, $(a \odot b)_i = a_i b_i$
$\langle a, b \rangle$	inner product $a^\top b = \sum_i a_i b_i$; $\|a\| = \sqrt{a^\top a}$
$\nabla_x F$	gradient of scalar F w.r.t. vector x — the vector of partials $\partial F / \partial x_i$
$\dot{x}$	time derivative dx/dt
$f'(\cdot)$	derivative of the scalar activation f, applied elementwise
$\mathbb{1}[\cdot]$	indicator: 1 if the condition holds, else 0
$\mathrm{sgn}(z)$	sign of z ($+1$ or -1)
$\mathrm{sg}[x]$	**stop-gradient**: equals x on the forward pass but $\partial\,\mathrm{sg}[x]/\partial x = 0$

INDICES

$i,\ j$	individual neurons — postsynaptic i, presynaptic j
l	layer (cortical hierarchy)
t	time
μ	stored memory pattern
k	learning module / system ($k = 1 \ldots 5$)

The universal skeleton, once more

MASTER RULE

$$\Delta w_{ij} \;=\; \eta\, \ell_{ij}\, m_i$$

w_{ij}	synaptic weight from presynaptic neuron j to postsynaptic neuron i
$\eta > 0$	learning rate (step size) — later made *time- and age-dependent*, $\eta(t)$
ℓ_{ij}	**local factor** — computable at the synapse from pre/post activity alone (Hebbian material)
m_i	**teaching factor** delivered to the postsynaptic cell — the learning signal that differs across systems

Three clocks on one synapse

The same w_{ij} is governed by processes that run orders of magnitude apart in time. Keeping them separate is what makes the whole picture coherent.

CLOCK	TIMESCALE	VARIABLE THAT MOVES	GOVERNED IN
Inference	$\sim$10–100 ms	activities x relax to $x^\star$	every section (fast dynamics)
Plasticity	seconds–hours	weights w within one brain state	§1–§5
Consolidation	hours–days	transfer + renormalisation w	§8–§9 (sleep)
Development	years–decades	which w_{ij} *exist*; $\eta(t)$	§10 (growth/pruning)

Space (which of the five systems) × fast time (wake/sleep) × slow time (lifespan) is the coordinate system for the whole document.

01 · CEREBELLUM

Supervised delta rule

The cleanest learner in the brain: one plastic layer, an explicit error wire, exact gradient descent. Clean precisely *because* it has no depth to assign credit through.

FORWARD MODEL

EXPANSION + READOUT

$$g = \sigma(Mu - \theta), \qquad y = w^\top g = \sum_{k=1}^{N} w_k g_k$$

TEACHING SIGNAL & LEARNING

CLIMBING-FIBRE ERROR

$$c = y - y^\star, \qquad \Delta w_k = -\eta\, c\, g_k$$

Because y is linear in w, this is *exact* gradient descent on the squared error $E = \frac{1}{2}(y - y^\star)^2$:

GRADIENT IDENTITY

$$\frac{\partial E}{\partial w_k} = (w^\top g - y^\star)\, g_k = c\, g_k \implies \Delta w = -\eta\, \nabla_w E$$

VARIABLES

$u \in \mathbb{R}^d$	mossy-fibre input; d input channels
$M \in \mathbb{R}^{N \times d}$	fixed sparse granule-cell input matrix (non-plastic)
θ	granule-cell thresholds; σ the threshold nonlinearity
$g \in \mathbb{R}^N$	granule / parallel-fibre activity — a high-dimensional **sparse expansion** of u, $N \gg d$
$w \in \mathbb{R}^N$	parallel-fibre → Purkinje weights — **the only plastic parameter**
$y,\ y^\star$	Purkinje output (simple-spike rate) and its target
c	climbing-fibre error, delivered as complex spikes from the inferior olive: $m_i = -c$

CELLS, SYNAPSES & DENSITY

The circuit maps one-to-one onto the equation. Granule cells *are* the vector g; the Purkinje cell *is* the readout $w^\top g$; the climbing fibre *is* c.

ELEMENT	TRANSMITTER / TYPE	MAPS TO	COUNT / DENSITY (HUMAN)
Granule cell	glutamatergic (excit.)	g_k — features	~50–101 billion; ~4 dendritic claws each; densest layer in brain (~1.6×10^6 / mm^3)
Purkinje cell	GABAergic (inhib.) output	$y = w^\top g$	15–26 million; up to ~200,000 PF synapses each
Climbing fibre	glutamatergic, from inferior olive	c (teaching)	~1 : 1 onto Purkinje cells
PF→PC synapse	glutamatergic, AMPA; **LTD site**	plastic w_k	the learned weights

BRAIN'S NEURONS HERE	BRAIN MASS HERE	LEARNING DEPTH
~80%	~10%	1 layer

DIGITAL EQUIVALENT & TRIGGER

An injected supervised label, fired by a *thresholded event detector* (the olive emits a near-binary complex spike on mismatch).

```
# inferior olive = comparator emitting a complex spike
error = y - y_target
if abs(error) > theta_olive:        # TRIGGER: mismatch crosses threshold
    m = -error                     # signed teaching signal
    w += eta * g * m               # delta rule, one layer, no transport
```

WHY IT IS SETTLED

The teaching signal is a physically identified, recorded quantity (climbing-fibre activity), and there is no hidden layer, so there is no credit-assignment problem to solve. This row of the table has no question marks.

02 · BASAL GANGLIA

Reinforcement learning & dopamine

Reward-prediction error, discovered in a neuron: phasic dopamine *is* the temporal-difference error δ_t, broadcast as a single scalar that gates every synapse.

CRITIC, TEACHING SIGNAL, UPDATES

VALUE ESTIMATE

$$V(s) = w^\top \phi(s) = \sum_{j=1}^{n} w_j \, \phi_j(s)$$

TD ERROR = DOPAMINE

$$\delta_t = r_t + \gamma \, V(s_{t+1}) - V(s_t)$$

ELIGIBILITY + CRITIC

$$e_t = \gamma\lambda \, e_{t-1} + \phi(s_t), \qquad w \leftarrow w + \alpha \, \delta_t \, e_t$$

ACTOR (POLICY GRADIENT)

$$\theta \leftarrow \theta + \alpha_\theta \, \delta_t \, \nabla_\theta \log \pi_\theta(a_t \mid s_t)$$

VARIABLES

s_t	state at time t
$\phi(s) \in \mathbb{R}^n$	feature vector of the state — physically supplied by cortical/striatal representations
$w \in \mathbb{R}^n$	critic (value) weights
$V(s)$	estimated value — expected discounted future reward
r_t	reward received; $\gamma \in [0, 1)$ the discount factor

δ_t	**reward-prediction error**, encoded by phasic dopamine — a single scalar: $m_i = \delta_t$
e_t	eligibility trace — bridges the delay between action and reward; $\lambda \in [0, 1]$
$\pi_\theta(a \mid s)$	the policy (action probabilities); θ its parameters (dorsal striatum)

CELLS, SYNAPSES & DENSITY

The striatum is ~95% one cell type. Two subtypes form opposing pathways; dopamine sets the sign of their plasticity.

ELEMENT	TRANSMITTER / TYPE	MAPS TO	COMPOSITION
Medium spiny neuron (D1)	GABAergic, "direct / Go"	policy / value units	~½ of MSNs; → GPi/SNr
Medium spiny neuron (D2)	GABAergic, "indirect / NoGo"	policy / value units	~½ of MSNs; → GPe (~40% co-express)
MSNs together	GABAergic projection	ϕ, π	~**95%** of striatal neurons
Cholinergic interneuron (TAN)	acetylcholine, tonically active	local gating	~1–2%
Dopamine axon (SNc/VTA)	dopaminergic, modulatory	δ_t (teaching)	broad, arborised
Corticostriatal synapse	glutamatergic (AMPA/NMDA), DA-gated	learned w	three-factor plasticity site

DIGITAL EQUIVALENT & TRIGGER

A single global scalar (the `advantage` of actor–critic), fired by a phasic neuromodulator burst on reward or a reward-predicting cue.

```
delta = r + gamma*V(s_next) - V(s)          # TD error
# TRIGGER: reward or reward-predictive cue -> phasic dopamine burst/dip
e = gamma*lam*e + grad_V(s)                 # eligibility (solves timing)
w += alpha * delta * e                      # one scalar gates every synapse
```

WHY IT IS SETTLED

δ_t is a theoretical quantity (the RL temporal-difference error) that was *found in a neuron*. The teaching signal is identified and recorded — no question marks.

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03 · CORTEX

Predictive coding & prospective configuration

The only system with depth — so the only one with a real credit-assignment problem, and the only one whose teaching signal is *not yet physically pinned down*. This is where the open question lives.

3.1 Architecture and error

TOP-DOWN PREDICTION

$$\hat{x}_l = W_l\, f(x_{l+1}), \qquad \varepsilon_l = x_l - \hat{x}_l = x_l - W_l\, f(x_{l+1})$$

$x_l \in \mathbb{R}^{n_l}$	activity of layer l; the **dynamical variables** that relax during inference. Layer 0 = sensory input, layer L = top
$s,\ t$	sensory input clamps $x_0 = s$; t is the target for x_L (present only when goal-directed)
W_l	top-down (generative) weights predicting layer l from layer $l+1$
$f,\ f'$	activation function and its derivative (elementwise)

$\hat{x}_l$	top-down prediction of layer l
ε_l	**prediction error** at layer l — hypothesised to live in *apical dendrites* (§3.7). This is the contested teaching signal m_i

3.2 The energy function

VARIATIONAL FREE ENERGY

$$F_\beta = \frac{1}{2}\sum_{l=0}^{L-1} \varepsilon_l^\top \varepsilon_l \; + \; \frac{\beta}{2}\,\|x_L - t\|^2$$

F_β is the single scalar both fast and slow dynamics descend. β ≥ 0 is the **target-coupling strength** — how hard the target is imposed. This one knob separates backprop from prospective configuration.

3.3 Fast dynamics — inference over activities

Descend F_β in the activities. An interior x_l enters F_β twice — directly in ε_l, and through $f(x_l)$ inside ε_{l-1}. Differentiating gives the error-propagation term:

INFERENCE (FAST, τ_x)

$$\tau_x\,\dot{x}_l = -\varepsilon_l + f'(x_l) \odot \left(W_{l-1}^\top \varepsilon_{l-1}\right)$$

with top boundary $\tau_x \dot{x}_L = f'(x_L) \odot (W_{L-1}^\top \varepsilon_{L-1}) - \beta(x_L - t)$ and $x_0 = s$ fixed. Run to equilibrium $\dot{x} = 0$; call it $x_l^\star$, with $\varepsilon_l^\star = x_l^\star - W_l f(x_{l+1}^\star)$.

3.4 Slow dynamics — learning over weights

Descend F_β in the weights, evaluated at the equilibrium. W_l appears only in ε_l:

PLASTICITY (SLOW)

$$\frac{\partial F_\beta}{\partial W_l} = -\varepsilon_l\, f(x_{l+1})^\top \;\Longrightarrow\; \Delta W_l = \eta\, \varepsilon_l^\star\, f(x_{l+1}^\star)^\top$$

Skeleton match: $\ell_{ij} = f(x_{l+1,j}^\star)$ (presynaptic activity of the layer above), $m_i = \varepsilon_{l,i}^\star$ (equilibrium error). Local: postsynaptic error × presynaptic activity, no transported weights.

3.5 The β limit — backprop as a boundary case

Let x^{ff} be the feedforward equilibrium (relax with $\beta = 0$, no target), $\mathcal{L} = \frac{1}{2}\|x_L^{\text{ff}} - t\|^2$ the output loss, and $\delta_l = \partial\mathcal{L}/\partial x_l$ the backprop error.

THE WEDGE

$$\lim_{\beta\to 0} \Delta W_l^{\text{PC}} \parallel -\nabla_{W_l}\mathcal{L} \qquad \text{(exact backprop, up to a positive scalar)}$$

	$\beta \to 0$ (BACKPROP LIMIT)	β FINITE (PROSPECTIVE CONFIG.)
Hidden activity	$x^\star \to x^{\text{ff}}$ (barely moves)	$x^\star \neq x^{\text{ff}}$ (settles to target-consistent state)
Update uses	the feedforward pass	the settled state, computed *before* any weight change
Result	the exact gradient	departs from the gradient; less cross-association interference

The difference is $O(\beta)$. This same F_β unifies predictive coding, equilibrium propagation, and contrastive Hebbian learning: all are F_β-descent, equal to backprop at $\beta \to 0$, diverging as β grows.

3.6 Precision weighting

CONFIDENCE-SCALED

$$F_\beta = \frac{1}{2}\sum_l \varepsilon_l^\top \Sigma_l^{-1}\varepsilon_l + \frac{\beta}{2}(x_L - t)^\top \Sigma_L^{-1}(x_L - t)$$

`Σ_l` is the covariance of layer-l errors; `Σ_l⁻¹` is **precision** (confidence), down-weighting unreliable channels. This is the mathematical slot where attention and neuromodulation enter — and it is the same knob the waking brain turns (§7).

3.7 The dendritic implementation of ε_l

The physical claim: the abstract error is a compartment voltage. Top-down drive to the apical tuft, minus tuned inhibition, equals the error.

APICAL = ERROR (HYPOTHESIS)

$$v_i^{\mathrm{ap}} \propto u_i^{\mathrm{td}} - q_i \overset{?}{\propto} \varepsilon_{l,i}, \qquad \Delta w_{ij}^{\mathrm{ff}} = \eta\big(\phi(v_i^{\mathrm{som}}) - \phi(\hat{v}_i)\big)\, r_j$$

v_i^{som}	somatic membrane potential — driven by bottom-up (basal) input; the neuron's output
v_i^{ap}	apical dendritic potential — driven by top-down input; hypothesised $= \varepsilon_{l,i}$
u_i^{td}	top-down excitatory drive to the apical tuft (the prediction)
q_i	inhibitory drive onto the tuft from SOM/VIP interneurons, tuned to cancel the prediction
$\hat{v}_i$	the dendrite's prediction of the soma; ϕ the rate nonlinearity; r_j presynaptic basal rate

Balanced prediction ⇒ apical quiet ⇒ $\varepsilon \approx 0$; mismatch ⇒ apical depolarisation. In network form the apical steady-state voltage equals the credit-assignment signal, $v_l^{\mathrm{ap,steady}} \propto \delta_l$.

CELLS, SYNAPSES & DENSITY

ELEMENT	TRANSMITTER / TYPE	MAPS TO	COMPOSITION
Pyramidal neuron	glutamatergic; two-compartment	x_l (soma) & ε_l (apical)	~**80%** of cortical neurons
PV basket interneuron	GABAergic, perisomatic	gain / f'-control	fast-spiking
SOM (Martinotti)	GABAergic, apical/dendritic	tuned inhibition q	targets the tuft
VIP interneuron	GABAergic, disinhibitory	gates q (opens learning)	inhibits SOM
Basal (feedforward) synapse	glutamatergic AMPA/NMDA	learned W	plasticity site

Apical (top-down) synapse	glutamatergic, layer-1	predictions $\hat{x}_l$	carries feedback

CORTICAL NEURONS	CORTICAL SYNAPSES	SYNAPSE DENSITY
~16–26B	~1.4×10^{14}	~10^9/mm^3

DIGITAL EQUIVALENT & TRIGGER

Decoupled local learning: every unit computes its own tiny loss from a *second input port* and updates on it — fired by a multiplicative coincidence gate (active × surprised).

```
# each unit has a SECOND input port (apical) carrying a top-down target
prediction = W_top_down @ f(x_above)
mismatch   = x_local - prediction              # local error, stop-gradient
gate = is_active(soma) and is_mismatched(apical)   # TRIGGER: coincidence (bu
if gate:
    w += eta * f(x_above) * mismatch           # learn only when active AND w
```

THE THREE OPEN IDENTITIES

This is the one row of the master table with question marks. **(i)** Does v^{ap} really track ε_l with the right functional form? **(ii)** Do synapses move by a dendrite-gated three-factor rule in vivo? **(iii)** Is $\beta \to 0$ (exact gradient) or β finite (prospective configuration)? Each of these has recently been tested against in-vivo data — the current answers, with citations, are laid out in **§11.3**. The short version: apical dendrites do carry vectorised error signals, plasticity is plateau/burst-gated with eligibility traces, and the cortex looks like finite-β prospective configuration rather than exact backprop.

04 · HIPPOCAMPUS

One-shot attractor memory

A different mathematics entirely: not gradient descent but a single-presentation Hebbian write into a recurrent net whose energy minima *are* the memories. The fast system — rate-1 storage, not small steps.

STORAGE, RECALL, ENERGY

COVARIANCE HEBB (ONE SHOT)

$$W_{ij} = \frac{1}{N} \sum_{\mu=1}^{P} \xi_i^\mu \, \xi_j^\mu, \qquad W_{ii} = 0$$

RECALL (ASYNCHRONOUS)

$$s_i \leftarrow \operatorname{sgn}\Big(\sum_j W_{ij} s_j \Big)$$

LYAPUNOV ENERGY (NON-INCREASING)

$$E(s) = -\tfrac{1}{2} \sum_{i,j} W_{ij} s_i s_j = -\tfrac{1}{2}\, s^\top W s$$

VARIABLES

$\xi^\mu \in \{-1,+1\}^N$	the μ-th stored pattern; the target state of each neuron
$P,\ N$	number of stored patterns; number of neurons
W_{ij}	recurrent weight $j \to i$ — the autoassociative memory matrix
$s \in \{-1,+1\}^N$	current network state during recall
$E(s)$	energy; its local minima are the stored patterns; recall descends it (pattern completion)

Skeleton match: $\ell_{ij} = \xi_j^\mu$, $m_i = \xi_i^\mu$ — **the target is the pattern itself** (self-supervised), learning rate $\eta = 1/N$, one presentation. Capacity $P_{\max} \approx 0.138\, N$; above it, spurious minima proliferate.

PATTERN SEPARATION (DENTATE GYRUS)

SPARSIFY BEFORE STORING

$$\Psi : \xi \mapsto \tilde{\xi}, \qquad a = \frac{\|\tilde{\xi}\|_0}{N} \ll 1$$

Ψ is a sparsifying pre-map orthogonalising patterns to raise effective capacity; a is the coding sparsity; $\|\cdot\|_0$ counts nonzero entries.

CELLS, SYNAPSES & DENSITY

ELEMENT	TRANSMITTER / TYPE	MAPS TO	COUNT / ROLE (HUMAN)
DG granule cell	glutamatergic, very sparse	separation map Ψ	~15 million; extreme sparsity
CA3 pyramidal	glutamatergic, recurrent	**the Hopfield W**	~2.7 million; dense recurrent collaterals
CA1 pyramidal	glutamatergic, output	comparator / readout	~14 million
Mossy fibre DG→CA3	glutamatergic "detonator"	write pattern into W	sparse, powerful
CA3→CA3 recurrent	glutamatergic NMDA, plastic	W_{ij}	autoassociation site

DIGITAL EQUIVALENT & TRIGGER

A content-addressable / key–value memory with a write-enable gated by novelty and theta phase.

```
if novelty(pattern) > theta_nov and phase == ENCODE:  # TRIGGER: novel + righ
    W += outer(pattern, pattern) / N       # one-shot write, rate = 1
s = sign(W @ cue)                          # energy descent -> completion
```

THE FAST HALF OF A TWO-SPEED MEMORY

Hippocampus writes in one shot; cortex (§3) learns slowly. Their relationship — replay-buffer to slow-learner — is the coupling that consolidation implements (§8). This is Complementary Learning Systems in two equations.

05 · NEUROMODULATORY GLUE

Three-factor gating

Not a separate module but the *skeleton itself* made literal: local coincidence lays down a trace; a global chemical signal decides whether and how it becomes a weight change. Choosing that signal reproduces every system above.

ELIGIBILITY TRACE AND GATED UPDATE

THE GENERAL FORM

$$\dot{e}_{ij} = -\frac{e_{ij}}{\tau_e} + H\big(x_i^{\text{post}}, x_j^{\text{pre}}\big), \qquad \Delta w_{ij} = \eta\, M(t)\, e_{ij}$$

e_{ij}	eligibility trace — a decaying memory of recent pre–post coincidence; τ_e its time constant
$H(\cdot,\cdot)$	Hebbian/STDP coincidence kernel (e.g. $x_i^{\text{post}} x_j^{\text{pre}}$)
$M(t)$	the global neuromodulatory signal — the **third factor**: dopamine, acetylcholine, or noradrenaline

Setting $M(t) = \delta_t$ and $H = \phi_j$ reproduces §2 exactly. This is the master skeleton; §1–§4 are choices of what fires the gate.

THE FOUR NEUROMODULATORS AS ONE LEVER EACH

SYSTEM	SOURCE (HUMAN COUNT)	SIGNAL M	WHAT IT SETS IN THE MODEL
Dopamine	SNc + VTA (~0.4–0.6 M)	δ_t (RPE)	reward-gated sign of plasticity
Acetylcholine	basal forebrain / NBM (~0.2 M)	encode ↔ consolidate switch	apical/feedback gain g_{ap}, β; ↑ = encode

Noradrenaline	locus coeruleus (∼ 50 k)	arousal / gain / "protect"	gain (precision Σ^{-1}); tags synapses against downscaling
Serotonin	raphe (∼0.2–0.3 M)	tonic state / patience	discount γ, overall tone

Each nucleus is tiny but projects via massively arborised axons (one neuron → up to ∼ 10^5–10^6 varicosities) using volume transmission — the anatomical meaning of "one scalar broadcast to a whole population."

DIGITAL EQUIVALENT & TRIGGER

A global gate on a temporal buffer, fired by surprise / reward / uncertainty detectors (VTA, basal forebrain, locus coeruleus).

```
e = decay*e + hebb(pre, post)     # eligibility buffer accumulates coincidenc
# TRIGGER: VTA / basal forebrain / locus coeruleus fire on reward, novelty, u
w += eta * M * e               # M in {dopamine, ACh, NE} flushes the buffer
```

THE UNIFYING INSIGHT

Four delivery modes for the teaching signal m: *label injection* (cerebellum), *scalar broadcast* (basal ganglia, neuromod), *per-unit local computation* (cortex), *identity target* (hippocampus). Every trigger is a gate — threshold, phasic broadcast, coincidence-AND, or novelty write-enable. None uses transported gradients.

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06 · COUPLING

The gradient-cut architecture

The five systems pass *activations* to each other through the thalamus and cortico-cortical loops, but no gradient crosses a module boundary. There is no global loss. In machine-learning terms: `stop_gradient()` on every inter-module edge, each module trained by its own head.

THE TWO DEFINING EQUATIONS

FORWARD COUPLING

$$x^{(k)} = F_k\big(x^{(k)},\ \{\mathrm{sg}[x^{(j)}]\}_{j\in \mathrm{in}(k)};\ \theta_k\big)$$

GRADIENT ISOLATION

$$\frac{\partial \mathcal{L}_k}{\partial \theta_j} = 0 \quad (j \neq k), \qquad \Delta\theta_k = -\eta_k \frac{\partial \mathcal{L}_k}{\partial \theta_k}$$

$k \in \{1, \ldots, 5\}$	module index (the five systems)
$\theta_k,\ x^{(k)}$	parameters and output/activity of module k
$\mathcal{L}_k$	the **local** objective of module k, driven by its own teaching signal $m^{(k)}$
F_k	forward function; in(k) the modules feeding into k (largely via thalamus)
$\mathrm{sg}[\cdot]$	stop-gradient — forward-identity, zero-derivative. This *is* the "gradient cut"

The system jointly-but-independently minimises $\{\mathcal{L}_1, \ldots, \mathcal{L}_5\}$ — *not* a single $\mathcal{L} = \sum_k \mathcal{L}_k$ with shared gradients. Activations cross boundaries; gradients do not.

THE CONCRETE COUPLINGS, WITH THEIR ML NAMES

PAIR	ANATOMY	WHAT CROSSES	ML ANALOGUE
Cortex → basal ganglia	corticostriatal; return via thalamus	features $\phi(s)$; $\partial\mathcal{L}_{\mathrm{BG}}/\partial\theta_{\mathrm{ctx}} = 0$	frozen backbone + actor–critic head
Hippocampus → cortex	replay during rest/sleep	replayed samples $\tilde{x}$, not gradients	experience-replay buffer + distillation
Cerebellum ↔ cortex	efference copy; return via thalamus	corrective outputs	bolt-on forward model / supervised corrector
Arbitration	prefrontal + basal ganglia	control weight b_k	mixture-of-experts router

THE ROUTER

UNCERTAINTY-BASED ARBITRATION

$$b_k(t) = \frac{1/\mathrm{unc}_k(t)}{\sum_{k'} 1/\mathrm{unc}_{k'}(t)}, \qquad \text{behaviour} = \sum_k b_k(t)\, x^{(k)}$$

`b_k(t)∈[0,1]` is how much module k drives behaviour ($\sum_k b_k = 1$); `unc_k(t)` its estimated unreliability. Control flows to whichever system is currently most reliable — goal-directed vs habitual, place vs response, model-based vs model-free.

HIPPOCAMPUS→CORTEX, MADE EXPLICIT

REPLAY = TRAINING DATA

$$\tilde{x} \sim p_{\mathrm{HPC}}(\cdot), \qquad \theta_{\mathrm{ctx}} \leftarrow \theta_{\mathrm{ctx}} - \eta\, \nabla_{\theta_{\mathrm{ctx}}} \mathcal{L}_{\mathrm{ctx}}(\tilde{x})$$

WHY ISOLATE THE GRADIENTS AT ALL

Gradient isolation *is* the brain's solution to global credit assignment. Transporting one gradient across the whole brain would resurrect weight-transport at planetary scale. Instead the brain factorises into modules with local teaching signals, coupled only by activations and arbitrated by a router — buying robustness, independent timescales, and independent signals, at the cost of never being globally optimal. Which is exactly why cortex (§3) is a finite-β *approximate* gradient rather than an exact one: a system that will not transport a clean gradient for its own loss is certainly not doing exact end-to-end optimisation across five of them.

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07 · THE WAKING BRAIN

Wake — encode, act, accumulate

Wake is not a single state. A neuromodulatory tone vector sets the gains, gates, and learning mode of all five systems at once; different wake phases turn that lever to "encode." The net effect is potentiation — the brain accrues a debt it will repay in sleep (§8).

7.1 The neuromodulatory state vector

All of wake vs sleep, and the phases within each, are captured by four scalars that the §5 nuclei broadcast:

BRAIN-STATE CONTROL

$$\mathbf{M}(t) = \big(a_{\mathrm{ACh}},\, a_{\mathrm{NE}},\, a_{\mathrm{DA}},\, a_{\mathrm{5HT}}\big)$$

a_{ACh}	acetylcholine — sets the **encode** ↔ **consolidate** switch and attention
a_{NE}	noradrenaline — arousal, gain, precision, and synaptic "protect" tagging
a_{DA}	dopamine — phasic reward-prediction error (the δ_t of §2)
a_{5HT}	serotonin — tonic state, patience, effective discount γ

7.2 How the state sets the machinery

The key lever is acetylcholine acting on feedback. High ACh *suppresses* top-down/recurrent input relative to feedforward drive — biasing the cortex toward registering new sensory input rather than replaying stored predictions (the Hasselmo principle).

GAIN GATING

$$g_{\mathrm{ap}}(t) = g^{0}_{\mathrm{ap}}\,\sigma\big(-\kappa_A\, a_{\mathrm{ACh}}\big), \qquad \Sigma^{-1}(t) \;\propto\; a_{\mathrm{NE}}$$

g_{ap}	effective apical (top-down / error-feedback) gain — the weight on ε_l in §3.3. **Low in waking** (high ACh) ⇒ encoding mode; high in NREM ⇒ replay/consolidation mode
κ_A	coupling constant; σ a squashing nonlinearity
Σ^{-1}	precision from §3.6 — arousal (NE) sharpens which errors count

So the single β/precision knob of the cortical model is *physically the neuromodulatory state*. Wake and sleep are two settings of the same equation.

7.3 Wake phases

PHASE	TONE $\mathbf{M}$	DOMINANT FLOW	LEARNING OPERATION
Active / attentive	ACh↑ NE↑ (phasic DA)	feedforward, sensory-driven	encode; accumulate eligibility; RL updates on reward
Exploratory (theta)	ACh↑, hippocampal θ 4–8 Hz	bottom-up storage interleaved with recall	phase-separated encode/retrieve (7.4)
Quiet / restful	ACh↓ NE↓	awake sharp-wave ripples	micro-consolidation; awake replay

7.4 The theta phase code — encode without overwriting

Within each hippocampal theta cycle, storage and recall are separated in time so that writing a new memory does not corrupt reading an old one:

PHASE-GATED PLASTICITY

$$\eta_{\text{HPC}}(t) = \eta_0 \, \mathbb{1}\left[\theta_{\text{phase}}(t) \in \text{trough}\right], \qquad \text{retrieval when } \theta_{\text{phase}} \in \text{peak}$$

Encoding is enabled only on the trough half-cycle (strong feedforward, plasticity on); the peak half-cycle reads out existing attractors with plasticity off. A single $\sim$125 ms cycle time-shares the storage and recall operations of §4.

7.5 What each channel carries

The question "what type of information can pass through each variable" has a precise answer: channels are typed by *bandwidth* and *content*, and gated by $\mathbf{M}$.

CHANNEL / VARIABLE	PHYSICAL SUBSTRATE	INFORMATION CARRIED	BANDWIDTH
$x_{l-1} \to$ basal	feedforward glutamatergic	sensory content, bottom-up evidence	high-dim vector

$\hat{x}_l \to$ apical	top-down glutamatergic (L1)	predictions, context, expectation	high-dim vector (ACh-gated)
ε_l	apical potential v^{ap}	prediction error — "how wrong, and which way"	high-dim vector
δ_t	dopamine (volume)	reward / salience surprise	**one scalar**
a_{NE}	noradrenaline (volume)	arousal, uncertainty, network reset	one scalar (quasi-global)
a_{ACh}	acetylcholine (volume)	mode: encode vs consolidate; attention	one scalar / regional
e_{ij}	synaptic eligibility (molecular)	which pairs recently co-fired — a temporal bridge	per-synapse trace

The design principle is visible: content is vectorial and fast; teaching and mode are scalar and slow. High-dimensional messages ride glutamatergic wires; one-dimensional gates ride the neuromodulators.

7.6 The accumulating debt

Across waking, encoding is LTP-dominated, so the mean weight change is positive and a homeostatic load builds — the quantity that predicts the depth of the coming night's slow-wave activity.

NET WAKE DRIFT + SLEEP PRESSURE

$$\langle \Delta w \rangle_{\mathrm{wake}} > 0, \qquad \frac{dL}{dt}\bigg|_{\mathrm{wake}} = +\kappa \, \langle \ell_{ij} \, m_i \rangle$$

$L(t)$	synaptic load / homeostatic pressure — rises with cumulative potentiation, discharged in NREM (§8). Correlates with EEG slow-wave activity
κ	accumulation constant

WHY WAKE CANNOT GO ON FOREVER

Continued LTP saturates the dynamic range of synapses and raises noise, degrading signal-to-noise and exhausting the capacity for further LTP. The rising $L(t)$ is the mathematical statement of "sleep need." Sleep is the reset.

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08 · THE SLEEPING BRAIN

Sleep — replay, transfer, renormalise, prune

Sleep is the offline half of the algorithm. NREM runs two operations at once — a gated hippocampus → cortex *transfer* and a global *downscaling*; REM does integration and structural pruning. Together they discharge the wake debt and convert fragile traces into stable cortical structure.

8.1 Sleep architecture

Human sleep cycles through stages roughly every 90 min, 4–6 times a night. Slow-wave (N3) dominates early cycles; REM lengthens toward morning.

STAGE	TONE $\mathcal{M}$	SIGNATURE RHYTHM	OPERATION
N1 (light)	ACh↓ NE↓	theta, drift	transition
N2	ACh↓ NE↓	spindles, K-complexes	onset of transfer + protection
N3 (SWS)	ACh↓↓ NE↓	SO ⊥ spindle ⊥ ripple	transfer + downscaling (8.3–8.4)
REM	ACh↑ NE≈0 5HT≈0	theta 4–8 Hz, PGO	integration + pruning (8.5)

8.2 The three nested rhythms

NREM consolidation is carried by three oscillations at very different frequencies, phase-locked into a hierarchy:

RHYTHM	BAND	ORIGIN	ROLE IN THE MODEL
Slow oscillation $\mathrm{SO}(t)$	0.16–4 Hz	neocortex (UP/DOWN)	global window of excitability
Spindle $\sigma(t)$	8–16 Hz	thalamocortical	opens plasticity; relays cortical reactivation
Sharp-wave ripple $\rho(t)$	80–300 Hz	hippocampus CA1	emits the replayed memory pattern

The coupling is a phase-amplitude hierarchy: the SO up-state groups spindles; the spindle troughs group ripples. Define the **consolidation gate** as their alignment:

TRIPLE-COUPLING GATE

$$G(t) = \underbrace{\mathrm{SO}_{\uparrow}(t)}_{\text{up-state}} \cdot \underbrace{\sigma(t)}_{\text{spindle}} \cdot \underbrace{\rho(t)}_{\text{ripple}} \in [0,1]$$

`G(t)` is nonzero only in the brief windows where all three align — the moments when hippocampal output can write to cortex. It is the temporal key that gates every consolidation update below.

8.3 Replay-gated systems consolidation

During ripples, CA3/CA1 reactivate stored patterns — in compressed, often time-reversed sequences, biased toward recently-encoded and reward-tagged experiences. Each replayed pattern p_t becomes a training sample for cortex, applied only when the gate is open:

GATED TRANSFER (CORTEX LEARNS)

$$p_t \sim p_{\mathrm{HPC}}(\cdot\,;\text{recency, salience}), \qquad \Delta W_{\mathrm{ctx}}(t) = \eta_c\, G(t)\, \varepsilon_{\mathrm{ctx}}(p_t)\, f(x_{\mathrm{above}})^{\top}$$

This is exactly the §3 cortical rule and the §6 replay coupling, now time-gated by $G(t)$. Over many nights the memory *migrates*: the hippocampal trace is released while the cortical one is built.

HAND-OFF DYNAMICS

$$\frac{dW_{\mathrm{HPC}}}{dt} = -\lambda_H\, W_{\mathrm{HPC}}, \qquad \frac{dW_{\mathrm{ctx}}}{dt} = +\eta_c \left\langle G\, \varepsilon_{\mathrm{ctx}}\, f^{\top} \right\rangle$$

p_{HPC}	the replay distribution — a prioritised sample of stored patterns (recent, salient, tagged)
η_c	consolidation learning rate (slow)
λ_H	hippocampal release rate — the trace fades as cortex absorbs it

8.4 Homeostatic downscaling (SHY)

Concurrently, NREM renormalises synaptic strength globally, undoing the wake potentiation of §7.6 while *preserving relative differences*. The canonical form is multiplicative, so ratios survive:

SELECTIVE DOWNSCALING

$$\frac{dw_{ij}}{dt}\bigg|_{\text{NREM}} = -\gamma_{\downarrow}\, w_{ij}\left(1 - P_{ij}^{\text{prot}}\right), \qquad P_{ij}^{\text{prot}} = \mathbb{1}\left[\text{replayed / NE-tagged}\right]$$

$\gamma_{\downarrow}$	downscaling rate; acts on all synapses…
P_{ij}^{prot}	…except those **protected** — recently replayed (8.3) or noradrenaline-tagged during wake (molecularly, Homer1a / Arc)

The net 24-hour operation is **potentiate then renormalise**. Its two effects combine constructively:

NET EFFECT ON A MEMORY

$$\underbrace{+\,\eta_c G\,\varepsilon}_{\text{strengthen tagged}} \quad \underbrace{-\,\gamma_{\downarrow} w\,(1 - P^{\text{prot}})}_{\text{weaken the rest}} \implies \uparrow \text{signal-to-noise}$$

Relative strengthening of what matters, absolute weakening of the background: saturation is avoided, noise is cut, and capacity for tomorrow's learning is restored. This reconciles the two great sleep theories — active replay (strengthen) and synaptic homeostasis (downscale) — as the two terms of one update.

8.5 REM — integration and pruning

REM is chemically singular: acetylcholine returns to waking levels while noradrenaline and serotonin fall silent. The loss of the aminergic "protect" signal opens a window for depotentiation and structural elimination; theta supports cross-memory integration.

USE-IT-OR-LOSE-IT PRUNING

$$P_{\text{prune}}(w_{ij}) = P_0\, e^{-w_{ij}/w_c} \quad \text{(applied in REM, no protect signal)}$$

SCHEMA FORMATION (ABSTRACTION)

$$W_{\text{ctx}} \leftarrow W_{\text{ctx}} - \eta_s\, \nabla_W \sum_{\mu \in \text{replayed}} \left\| W - \text{shared}(\{p^\mu\}) \right\|^2$$

w_c	protection scale — synapses weaker than w_c are likely eliminated; strong ones survive
η_s	schema-integration rate; shared$(\cdot)$ extracts structure common across replayed memories

TWO KINDS OF FORGETTING, BOTH USEFUL

NREM downscaling (8.4) is *graded* — it shrinks weights while sparing the tagged. REM pruning (8.5) is *structural* — it removes weak synapses entirely and abstracts what remains into schemas. The first improves SNR; the second frees capacity and generalises. Both feed directly into the developmental pruning of §10 — the nightly mechanism is the lifelong mechanism, dialled up in youth.

The consolidation and downscaling equations in 8.3–8.5 are a formalisation compiled from the active-systems-consolidation and synaptic-homeostasis literatures — the biology fixes the *structure* (gated replay + global renormalisation + REM pruning); the exact rates are model parameters — though §12 shows most of them are not independent, closing instead to a homeostatic balance, an estimated precision, or an environmental statistic.

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09 · THE 24-HOUR CYCLE

Wake–sleep as one algorithm

The daily loop is a complementary-learning-systems training schedule: encode fast and cheaply while awake, then replay-transfer and renormalise while asleep. Written as a procedure, it is a two-phase optimisation over two networks.

THE DAILY LOOP

```
while alive:

  #  ---- WAKE : high ACh, high NE  (encode mode, g_ap low) ----
  for experience x in day:
      x_star = relax(x, beta_small)            # §3.3 fast inference (10-100
      W_hpc += outer(sep(x_star), sep(x_star)) # §4 one-shot store (fast, che
      e     = decay*e + hebb(pre, post)        # §5 accumulate eligibility
      W_k  += eta_k * e * m_k                   # §1,2 local updates on teach
      L    += kappa * potentiation             # §7.6 sleep debt rises

  #  ---- NREM : ACh low  (consolidate mode, g_ap high) ----
  while L > L_min:
      if G(t) == open:                        # SO ⊥ spindle ⊥ ripple aligned
          p = replay(W_hpc; recency, salience) # §8.3 prioritised sample
          W_ctx += eta_c * eps_ctx(p) * f_above # cortex learns the trace
          W_hpc -= lam_H * W_hpc               # hippocampus releases it
      W     *= (1 - gamma_down*(1 - protect))  # §8.4 global downscale (spare
      L     -= discharge                        # debt repaid

  #  ---- REM : ACh high, NE=0, 5HT=0  (integrate + prune) ----
  prune(W, p_prune ~ exp(-w/w_c))              # §8.5 remove weak synapses
  W_ctx = integrate_schemas(replayed)          # abstract shared structure
```

THE HOMEOSTATIC FIXED POINT

A stable brain is one where each day's potentiation is matched by that night's renormalisation — otherwise weights run away or collapse:

DAILY BALANCE

$$\underbrace{\langle \Delta w \rangle_{\text{wake}}}_{>0,\ \text{LTP}} + \underbrace{\langle \Delta w \rangle_{\text{sleep}}}_{<0,\ \text{downscale}} \approx 0 \qquad \text{(with SNR strictly improved)}$$

The equality is homeostatic; the inequality (SNR up) is what makes it *learning* rather than mere forgetting. Sleep loss breaks the balance: L never discharges, synapses saturate, and both new encoding and old-memory SNR degrade — the mathematical signature of sleep deprivation.

WHY TWO NETWORKS, TWO PHASES

One fast writer (hippocampus) prevents catastrophic interference by storing each experience separately and instantly; one slow integrator (cortex) extracts statistical structure by learning from replayed samples over many nights. Interleaving them offline is how the brain gets both one-shot memory *and* generalisation from the same experiences — a solution modern continual-learning systems now borrow explicitly.

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10 · DEVELOPMENT

The lifespan clock — grow, prune, stabilise

The slowest clock rewrites the substrate itself. The brain over-builds connections in childhood, selectively eliminates them through adolescence, and settles into slow turnover in adulthood. The learning rate is not a constant — it is a lifespan envelope, staggered across regions and systems.

10.1 Three phases

PHASE	AGE (HUMAN)	STRUCTURAL OPERATION	LEARNING REGIME
Child	0 – ~11	**overproduction** — synaptogenesis (∼40,000/sec at GA wk 34)	critical periods open; η high; high plasticity
Pre-adult / adolescent	~11 – early 20s	**pruning** — LTD-tagged elimination, esp. PFC	windows close; η falls; stability rises
Adult	20s +	slow turnover; ∼85,000 cortical neurons lost/day	low but non-zero plasticity

10.2 The synaptic-density trajectory

Density overshoots the adult level, then is pruned back. The peak and its timing are region-specific: sensory cortex early, association and prefrontal cortex late.

VISUAL CORTEX PEAK	PFC PEAK	PEAK → ADULT
140–150% of adult, 4–12 mo	**~1–3 yr** prunes into 20s	**~ −40%** synapses removed

GROWTH–PRUNING DYNAMICS

$$\frac{dN_r}{dt} = \underbrace{\alpha_g(t)\, N_r \left(1 - \frac{N_r}{N_{\max}}\right)}_{\text{synaptogenesis}} - \underbrace{\beta_p(t)\, N_r \left\langle 1 - w/w_c \right\rangle}_{\text{activity-dependent pruning}}$$

$N_r(t)$	number of synapses in region r at age t
$\alpha_g(t)$	growth rate — large in childhood, decaying with age
$\beta_p(t)$	pruning rate — rises in adolescence (region-specific onset)
$N_{\max}$	carrying capacity of the local circuit

10.3 Pruning is the nightly rule, slowed down

Which synapses are eliminated is activity-dependent — weak and unused connections go, strong ones stay (microglia- and complement-mediated). The functional form is *identical* to REM pruning (§8.5):

USE-IT-OR-LOSE-IT (SAME AS §8.5)

$$P_{\text{prune}}(w_{ij}) = P_0\, e^{-w_{ij}/w_c}$$

This is a genuine unification: the mechanism that trims a memory each night is the mechanism that sculpts the cortex across two decades — the same selection rule, applied at different rates. Sleep is developmental pruning's daily engine, which is why sleep is deepest and most abundant in the young, and why the consolidation rhythms of §8 are themselves still maturing in childhood (precise spindle–ripple coupling appears late).

10.4 The critical-period envelope

The learning rate itself rises and falls over the lifespan, gated shut by the maturation of perineuronal nets around fast-spiking (PV) interneurons:

AGE- AND REGION-DEPENDENT RATE

$$\eta_r(t) = \frac{\eta_{\max}}{1 + (t/t_c^r)^p}\,\big(1 - \mathrm{PNN}_r(t)\big), \qquad t_c^{V1} < t_c^{\text{assoc}} < t_c^{\text{PFC}}$$

$\eta_r(t)$	effective plasticity of region r at age t — the η of the master rule, now a lifespan function
t_c^r	critical-period midpoint for region r — early for V1, late for PFC (the staggering)
$\mathrm{PNN}_r(t)$	perineuronal-net maturation, rising to 1 and locking in the circuit (closes the window)
p	steepness of the closing

10.5 A parameter space that changes size

Because synapses are created and destroyed, the very set of learnable parameters is time-varying. The learning problem of §0 is solved over a space whose dimension itself follows the density curve:

STRUCTURAL LEARNING

$$\theta(t) \in \mathbb{R}^{|S(t)|}, \qquad |S(t)| \text{ tracks } N(t) \text{ (grow, then prune)}$$

`S(t)` is the set of existing synapses at age t. Overproduction supplies raw connectivity (structural exploration); activity-dependent pruning keeps what carries signal (structural exploitation). It is an explore-then-exploit search in *architecture space*, wrapped around the weight-space learning of §1–§5.

10.6 How the systems mature on different schedules

The staggering of t_c^r means the five-system architecture is assembled progressively, and the "adult" full system is the *last* stage, not the default one:

SYSTEM	MATURATION	CONSEQUENCE
Cerebellum / sensory cortex	early (infancy)	motor calibration and perception come online first
Hippocampus	early–mid childhood	episodic memory and its replay machinery mature; "infantile amnesia" precedes it
Basal ganglia	childhood	habit and reward learning functional early
Cortex (association / PFC)	late — into the 20s	prospective-configuration depth, abstraction
Router (PFC / arbitration)	latest	executive control, model-based over habitual — matures last

WHY PROLONGED IMMATURITY IS THE DESIGN, NOT A BUG

Late prefrontal pruning keeps the cortex plastic through the years of richest social and linguistic learning; the router matures last so that flexible, model-based control is layered on top of already-trained habit and perception systems. The extended human trajectory — synaptogenesis peaking in mid-childhood, pruning running into the third decade — is longer than in other primates, and buys exactly the deep, culturally-transmitted learning the cortex specialises in.

11 · THE FULL PICTURE

One system, three timescales

Assembled, the brain is a modular, multi-timescale learner: five local rules, coupled by activation but cut off from each other's gradients, gated hour-by-hour by neuromodulation into encode/consolidate phases, and rewired decade-by-decade by growth and pruning. No global loss, no transported gradient, no single optimiser — and yet it learns better than any of them.

11.1 The master table

Every rule in this guide is $\Delta w_{ij} = \eta\, \ell_{ij}\, m_i$. Here is what fills the three slots in each system, and how settled each row is.

SYSTEM	LOCAL ℓ_{ij}	TEACHING m_i	DELIVERED BY	STATUS
Cerebellum	g_j	$-(y_i - y_i^\star)$	climbing fibre	settled
Basal ganglia	$e_{t,j}$	$\delta_t = r_t + \gamma V' - V$	dopamine (scalar)	settled
Cortex	$f(x^\star_{l+1,j})$	$\varepsilon^\star_{l,i}$	apical dendrite v^{ap}	**converging — §11.3**
Hippocampus	ξ_j^μ	ξ_i^μ	the pattern (self-target)	settled
Neuromod. glue	e_{ij}	$M(t)$	dopamine / ACh / NE	settled (framework)

Coupled by $x^{(k)} = F_k(\dots, \mathrm{sg}[x^{(j)}], \dots)$ with $\partial\mathcal{L}_k/\partial\theta_{j\neq k} = 0$; arbitrated by $b_k(t)$; scheduled by $\mathbf{M}(t)$ across wake/sleep; and scaled by $\eta_r(t)$ across the lifespan.

11.2 The three timescales in one frame

AXIS	VARIABLE THAT MOVES	SET BY	SECTIONS
Space (which system)	choice of ℓ, m and cell type	anatomy	§1–§6
Fast time (~24 h)	$\mathbf{M}(t)$: encode ↔ consolidate	neuromodulatory nuclei	§7–§9
Slow time (~decades)	$\eta_r(t)$ and $\lvert S(t)\rvert$	growth / pruning / PNN	§10

A single synapse w_{ij} is simultaneously: relaxing toward an inference equilibrium (ms), being nudged by its local rule (minutes), being renormalised or transferred tonight (hours), and living inside a connectivity that is itself being built or pruned (years). All four are the same weight under four clocks.

11.3 What the evidence now says (2024–2026)

The one row of the master table with question marks is the cortical teaching signal. Three identities were open when this framework was first sketched; recent in-vivo recordings and modelling have moved all three — none is fully closed, but each has turned from a bare hypothesis into a hypothesis with data behind it.

(I) DOES THE APICAL DENDRITE CARRY THE ERROR?

Increasingly supported — with a live question about sign. Two-photon imaging of layer-5 pyramidal neurons during a brain–computer-interface task shows their apical dendrites carrying *vectorised* instructive signals — tailored per neuron, and spatially separated from the feedforward stream — which is exactly the property credit assignment needs and which a single scalar teaching signal cannot provide (*Vectorized instructive signals in cortical dendrites*, Nature 2026). In sensory cortex, layer-5 apical tufts report prediction-error-like activity during learning, though there it reads out as an *unsigned* error/salience term that drives dendritic gain rather than a cleanly signed ε_l (L5 apical imaging, S1 go/no-go, 2021–2024). Spiking models with two-compartment pyramidal cells plus PV/SOM/VIP interneurons reproduce the observed layer-2/3 mismatch responses and generate *sign-specific* positive and negative errors through compartment-specific excitation–inhibition balance (compartmental V1 model, 2025). *Verdict:* the apical compartment does carry a vectorised, credit-related error signal, as the model requires; whether it encodes signed ε_l directly or an unsigned magnitude that gates plasticity is the remaining sub-question.

(II) DO SYNAPSES MOVE BY A DENDRITE-GATED THREE-FACTOR RULE IN VIVO?

Confirmed in structure; the best-measured form is BTSP. In hippocampus, a dendritic plateau potential acts as a per-neuron gating factor that converts presynaptic eligibility traces into one-shot, bidirectional weight change over a seconds-long window — behavioural-timescale synaptic plasticity (Bittner & Milstein et al., Science 2017; Magee & Grienberger 2020). That is precisely the eligibility-trace × dendritic-gate structure of §5 and §3.7, observed directly. In cortex, longitudinal single-synapse imaging during motor learning shows apical and basal dendrites obeying *distinct* plasticity rules in vivo — basal strengthening tracks coincidence with postsynaptic spikes, apical strengthening tracks local clustered coactivity (compartment-specific rules in vivo, 2025). Canonical spike-timing plasticity, by contrast, appears to have little influence on in-vivo dynamics next to these plateau/burst-gated rules. *Verdict:* the dendrite-gated, eligibility-based, compartment-specific rule is real; its measured form (plateau/burst-gated, one-shot-capable) sits in the Urbanczik–Senn / three-factor family rather than being a literal continuous version of it.

(III) BACKPROP LIMIT OR PROSPECTIVE CONFIGURATION?

Evidence favours finite β. When a network settles its activity toward a target-consistent state *before* updating weights, it reproduces patterns of neural activity and behaviour across many learning tasks that backpropagation does not, and it outperforms backprop precisely in the online, few-shot, continual, and interference-prone settings a biological organism actually faces — while matching backprop's accuracy on standard supervised problems, of which backprop is the $\beta \to 0$ limit (Song et al., Nature Neuroscience 2024). The advantage depends on feedback being *influential* — substantial backward weights — so the operative regime is finite and not necessarily tiny β. *Verdict:* the cortex most likely runs approximate, target-consistent credit assignment at finite β, refining the earlier guess of $0 < \beta \ll 1$ toward "finite β, with feedback strong enough to matter."

THE META-BET, UPDATED

Three of these were open hypotheses; they are now hypotheses with in-vivo support. The apical compartment carries vectorised error signals (i); plasticity is gated by dendritic plateau/burst events combined with eligibility traces (ii); and behaviour and neural activity look more like finite-β prospective configuration than exact backprop (iii). The overall bet stands and is strengthened: there is no brain-wide loss and no transported gradient — learning is factorised into modules with local, dendritically-computed teaching signals, coupled by activations, scheduled by neuromodulation, and reshaped by development. Backpropagation remains a single limiting case (§3.5), not the mechanism. What is still genuinely unsettled is now narrow: whether the apical error is signed or an unsigned gate, and how large β actually is.

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12 · CLOSING THE CONSTANTS

Where the numbers come from

The spec looks like it has hundreds of free knobs — every η, every rate, every threshold. It doesn't. Studying where each one comes from shows the count is an illusion of the notation: almost every constant is either forced by a consistency condition, a shadow of one estimated quantity, or a statistic of the environment. This section closes them — and is honest about the few that stay open.

12.1 The three classes — and the one object

Sort every free parameter by *how* it is fixed:

CLASS	HOW IT CLOSES	FREEDOM IT ADDS
A · Consequence	forced by a stability, normalisation, or fixed-point condition	none
B · Shadow of precision	a function of the estimated error-covariance Π	none beyond Π
C · Environmental	equals a statistic of the world; estimated online	supplied by data, not chosen

The pivot is a single object. Define the **precision** — the inverse covariance of the prediction errors:

THE MASTER STATISTIC

$$\Pi \;=\; \Sigma^{-1} \;=\; \left(\mathbb{E}[\varepsilon\varepsilon^{\top}]\right)^{-1}, \qquad \Sigma \leftarrow (1-\rho)\,\Sigma + \rho\,\varepsilon\varepsilon^{\top}$$

Π is *not* a hyperparameter — it is a running estimate of the system's own error statistics (ρ a slow averaging rate). The claim of this section: the learning rates, the target-coupling β, the router weights, and the neuromodulatory gains are all *the same object*, Π, seen from different sides. Close Π and they close together.

12.2 The closures

LEARNING RATE — INVERSE CURVATURE (CLASS B)

Gradient descent on a loss with curvature (Hessian) H is stable only for $\eta < 2/\lambda_{\max}(H)$ and fastest at $\eta^{\star} = 1/\lambda$. Precondition by the Fisher curvature — the natural gradient — and the residual rate is exactly one:

NATURAL GRADIENT

$$\Delta\theta = -\,\Pi\,\nabla_\theta L, \qquad \Pi = F^{-1}, \quad F = \mathbb{E}\left[\nabla_\theta \log p\,\nabla_\theta \log p^{\top}\right]$$

So η is not a number to pick — it is the metric Π. The learning rate has disappeared into precision. (Amari, natural gradient.)

THE SCHEDULE IS FORCED (CLASS A)

For a stochastic estimate to converge the steps must satisfy $\sum_t \eta_t = \infty,\;\; \sum_t \eta_t^2 < \infty$ (Robbins–Monro); the minimum-variance online mean uses $\eta_n = 1/(n+1)$ exactly. Writing $n \approx r_r\, t$ —

informative samples arriving at rate r_r in region r:

CRITICAL PERIOD = RUNNING AVERAGE

$$\eta_r(t) = \frac{1}{1 + r_r\, t} \quad \Longleftrightarrow \quad \frac{\eta_{\max}}{1 + (t/t_c^{\,r})^p} \ \text{ with } \ p = 1,\ t_c^{\,r} = 1/r_r$$

The exponent $p = 1$ is forced for stationary statistics ($p > 1$ only if the target drifts — the region must stay plastic longer); the midpoint $t_c^{\,r}$ is the inverse rate of informative input. The staggering $t_c^{V1} < t_c^{\text{PFC}}$ is then not a choice but a fact about the world — sensory evidence arrives dense and fast, prefrontal and social evidence slow. $\text{PNN}(t)$ is the biology implementing $1 - \eta_r(t)/\eta_{\max}$.

TARGET COUPLING — RELATIVE PRECISION (CLASS B)

β weighs the target term against the prediction terms in F_β. Balancing a prediction against evidence is, in Bayesian terms, exactly weighing them by their precisions:

RELATIVE PRECISION

$$\beta = \frac{\Pi_{\text{target}}}{\Pi_{\text{pred}}} = \frac{\sigma^2_{\text{pred}}}{\sigma^2_{\text{target}}}$$

High when the model is uncertain (trust the target, move activity a lot), low when confident. The backprop limit $\beta \to 0$ is then "an overconfident model"; finite β is the calibrated regime — recovering the empirical answer of §11.3 from first principles. (Synthesis, consistent with active inference; not proven for the specific β of Song et al.)

THE ROUTER — PRECISION-WEIGHTED FUSION (CLASS B)

The optimal combination of estimators with variances σ_k^2 is inverse-variance weighting — Bayesian cue combination:

ARBITRATION IS BAYES

$$b_k = \frac{\Pi_k}{\sum_{k'} \Pi_{k'}}, \qquad \Pi_k = 1/\text{unc}_k$$

The uncertainties unc_k are each module's own precision — so the router reads quantities the modules *already compute*. There is nothing extra to train. The "who trains the router" worry dissolves; what remains is calibration — are the Π_k honest? — not arbitration. (Ernst–Banks cue combination.)

HOMEOSTATIC RATE — FIXED BY BALANCE; THE FORM IS FORCED (CLASS A)

A stable set point requires each 24-hour cycle to net to zero, $\langle \Delta w \rangle_{\text{wake}} + \langle \Delta w \rangle_{\text{sleep}} = 0$. That is one equation in one unknown:

BALANCE FIXES THE RATE

$$\gamma_{\downarrow} = \frac{\kappa \, \langle \ell \, m \rangle \, T_{\text{wake}}}{\big\langle w \, (1 - P^{\text{prot}}) \big\rangle \, T_{\text{sleep}}}$$

And requiring renormalisation to *preserve learned structure* — leave every ratio w_i / w_j unchanged — forces the multiplicative form $w \to \alpha w$ uniquely; it is the only rescaling that does. Both the rate and the shape of downscaling are consequences, not choices; stability of the set point (it must attract) gives the sign — downscaling rises with load. (SHY; Turrigiano multiplicative scaling.)

PRUNING THRESHOLD — THE NOISE/ENERGY FLOOR (CLASS A/C)

Keep a synapse only if its signal beats synaptic noise plus its metabolic cost. Elimination is then a threshold-crossing against noise of scale w_c:

SURVIVAL VS THE NOISE FLOOR

$$P_{\text{keep}}(w) = 1 - e^{-w/w_c}, \qquad w_c \approx \sigma_{\text{noise}} + (\text{energy cost})$$

w_c is set by the synaptic signal-to-noise ratio and the energy budget, not chosen. The nightly-and-lifelong e^{-w/w_c} of §8.5 and §10.3 is the same physics at two rates. (Attwell–Laughlin energy budget; wiring economy.)

THE ACTIVATION — THE INPUT'S CDF (CLASS C)

The nonlinearity that transmits the most information for an input distribution $p(u)$ is its cumulative distribution — histogram equalisation:

INFOMAX FIXES THE NONLINEARITY

$$f(u) = \int_{-\infty}^{u} p(u') \, du'$$

So f is not a modelling choice; it is the shape of the input statistics. Different inputs give different f, all derived. (Laughlin, 1981.)

THE ENVIRONMENTAL CONSTANTS — CLOSED AS ESTIMATORS, VALUED BY THE WORLD (CLASS C)

τ_e $=$ the action $\to$ outcome latency the trace must bridge; read from experience

γ	$=$ per-step continuation probability $P(\text{not terminated})$ — hazard-rate discounting
λ	$=$ the TD bias–variance optimum, set by value reliability vs return variance
p_{HPC}	$\propto$ surprise $\times$ reward $\times$ recency — computed from the same error and reward signals

Each equals a specific statistic — closed in *form* — but its *value* requires the environment. This is the irreducible environmental residue: the equations are determined, the numbers are not, until a world supplies them.

12.3 The collapse

Counting the truly independent objects after closure:

PARAMETER	CLOSES TO	CLASS	INDEPENDENT?
η	inverse curvature $= \Pi$	B	no
$\eta(t)$, t_c, p	$1/(1 + r\,t)$ schedule	A	no
β	$\Pi_{\text{target}}/\Pi_{\text{pred}}$	B	no
b_k (router)	$\Pi_k / \sum \Pi$	B	no
Σ^{-1} (precision)	estimated error covariance	—	**the one object**
$\gamma_{\downarrow}$, α	homeostatic fixed point	A	no
w_c	noise / energy floor	A/C	no
f	input CDF	C	world
$\tau_e, \gamma, \lambda, p_{\text{HPC}}$	environmental statistics	C	world

The "hundreds of constants" collapse to essentially **one estimated quantity** — the precision Π — plus a set of consistency conditions that add no freedom, plus the statistics of the environment, which are data rather than parameters. That is the answer to *why* the values are what they are:

they emerge from a single principle — **track your own uncertainty and let it set every gain** — and the rest is forced.

Provenance: the individual closures are standard results — natural gradient (Amari), Robbins–Monro, infomax/CDF matching (Laughlin), inverse-variance cue combination (Ernst–Banks), multiplicative synaptic scaling (Turrigiano / SHY), energy-budget wiring economy (Attwell–Laughlin), hazard-rate discounting. The unification — that these are one object, precision — is this document's synthesis, not a theorem; it is a lens that makes the closures cohere, and it is falsifiable wherever a constant refuses to reduce to Π, a consistency condition, or an environmental statistic.

12.4 What still does not close

Honesty requires marking the residue precisely. Three things survive, and no amount of studying the equations removes them.

`1 · Values need a world`	Class C closes each environmental constant to a *form* — an estimator — but not a *number*. f, τ_e, γ, the region rates r_r, the replay priority: none has a value until the environment's statistics are supplied. The world is still the missing input; the closure turns "free parameters" into "quantities to be measured," which is progress, not a number.
`2 · Existence of the joint fixed point`	Π estimates the errors while it *controls* the learning that changes those errors — a self-referential loop. The closure writes the equations of the loop; it does not prove they have a stable solution. Whether the coupled, five-module, precision-controlled, wake–sleep-scheduled, growing-and-pruning system converges rather than oscillating or collapsing is an open fixed-point question — settled, if at

	all, by analysis or simulation, never by inspection.
3 · The empirically open values	Theory says β should be "calibrated (finite)"; §11.3 says the evidence agrees, but whether the cortex actually sits at the calibrated value, and whether the apical error is signed, remain experimental questions. The principled target is derived; the confirmation is not.

THE HONEST BOTTOM LINE

Studying where the constants come from does close almost all of them — not by computing numbers, but by showing they were never independent: they are one estimated quantity, a few stability conditions, and the statistics of a world. That is a stronger claim than "we tuned everything," because a system with almost no arbitrary knobs is exactly the kind of thing evolution could discover and hold. But two gaps are structural, not clerical: the model still needs a *world* to give its estimators values, and it still needs a *proof or a simulation* that the self-referential precision loop is stable. The equations are nearly closed; the model is not yet alive. That distance — between a determined system of equations and a running one — is the same distance between writing the chemistry of a cell and building one. §13 takes up each of these three gaps and shows what closing it actually requires.

---

13 · CLOSING THE RESIDUE

The last three gaps, made operational

§12.4 left three things that studying the equations could not close: values need a *world*, the self-referential precision loop has no *stability proof*, and two cortical numbers are *empirically open*. None of these dissolves into a formula — but each converts from a mystery into something concrete: a checkable condition, a stability theorem with explicit hypotheses, or a decisive experiment. The residue stops being conceptual and becomes operational.

13.1 Gap one — values, by self-calibration under stationarity

The environmental constants do not need to be *known*; they need to be *estimated*, and the architecture already contains the estimators. Each is a running readout of the system's own experience:

THE ESTIMATORS ARE ALREADY THERE

$$\hat{\tau}_e = \arg\max_{\Delta} \ \operatorname{corr}\big(\text{event}(t),\ \text{outcome}(t+\Delta)\big), \quad \hat{\gamma} = 1 - \widehat{P_{\text{terminate}}}, \quad f_t \to F_u(u) = \int_{-\infty}^{u} p_u$$

$\hat{\tau}_e$	eligibility timescale — the lag at which events best predict outcomes; read from the cross-correlation the trace already accumulates
$\hat{\gamma}$	discount — one minus the observed termination hazard (running average of "still going")
$f_t \to F_u$	the activation drifts to the input's cumulative distribution by homeostatic gain adaptation — infomax done *online*, not set by hand

These converge on one condition: the world must be **stationary on the estimator's timescale** — its statistics must drift slower than the estimator settles.

THE STATIONARITY CONDITION

$$\tau_{\text{env}} \gg \tau_{\text{converge}}$$

If it holds, the values are supplied by the system itself. If it fails — a non-stationary world — the fix is to *track* rather than average: constant-rate forgetting (fixed η instead of $1/t$), which trades final precision for adaptivity and is exactly the knob arousal (noradrenaline) turns up when the world becomes surprising. So even non-stationarity is handled, at a stated cost.

And "a world" is not an unbounded unknown. The environment need only supply three structural properties for the five rules to have something to consume:

Temporal predictability	nonzero autocorrelation, so prediction error ε carries information

`An evaluative signal`	a scalar reward/cost, so δ_t is defined
`Sensorimotor contingency`	actions change future inputs, so credit is assignable at all

Verdict: Gap one closes to a **condition plus a spec** — self-calibrating estimators that converge whenever $\tau_{\text{env}} \gg \tau_{\text{converge}}$, inside any data stream carrying those three properties. It does not hand you numbers in the abstract; it tells you exactly what the world must provide and shows the system reading its own values off that world. Embodiment, reduced to three requirements.

13.2 Gap two — stability, by timescale separation and the gradient cut

The hard one. Precision Π estimates the errors while *controlling* the learning that changes those errors — a loop that could diverge. Three mechanisms, already present in the architecture, make it converge; together they turn "no theorem" into a theorem with explicit, satisfied hypotheses.

(A) THE THREE CLOCKS DECOUPLE THE LOOP

Write the coupled dynamics with their time constants — inference fast, learning slow, precision slower:

ONE SYSTEM, SEPARATED RATES

$$\tau_x\,\dot{x} = -\nabla_x F_\beta, \qquad \tau_w\,\dot{w} = -\nabla_w F_\beta, \qquad \tau_\Pi\,\dot{\Pi} = \Sigma - \Pi^{-1}, \qquad \tau_x \ll \tau_w \ll \tau_\Pi$$

With $\epsilon = \tau_x/\tau_w \to 0$, singular perturbation (Tikhonov) makes the fast variable track its equilibrium $x^\star(w,\Pi)$ while the slow ones look frozen; the same again for $w^\star(\Pi)$. The loop is not simultaneous — it is a hierarchy, and each level sees the faster ones as settled and the slower ones as constant. **Two-timescale stochastic approximation** then gives convergence with probability one:

TWO-TIMESCALE CONVERGENCE (BORKAR)

If, for each frozen (w,Π), the fast x-dynamics have a globally asymptotically stable equilibrium $x^\star(w,\Pi)$; and for each frozen Π, the slow flow $\dot{w} = -\nabla_w F_\beta(x^\star, w)$ has a stable equilibrium $w^\star(\Pi)$; and the precision flow has a stable $\Pi^\star$ — then

$(x_t, w_t, \Pi_t) \to (x^\star, w^\star, \Pi^\star)$ almost surely. The three clocks of §0 are not incidental; they *are* the stability mechanism.

(B) STOP-GRADIENT MAKES THE WHOLE THING A DESCENT

Because no gradient crosses a module boundary (§6), each module minimises its own energy given the stop-gradient'd activities of the others. Their sum is then a joint Lyapunov function, decreased by block-coordinate descent:

THE GRADIENT CUT BUYS A LYAPUNOV FUNCTION

$$\mathbb{F} = \sum_k F_k, \qquad F_k \text{ minimised in block } k \text{ given } \mathrm{sg}[x^{(j \neq k)}] \implies \mathbb{F} \text{ non-increasing}$$

Block-coordinate descent on an $\mathbb{F}$ bounded below, each step exact or improving, converges to a coordinatewise minimum. With full gradient coupling there would be no such decomposition — so the "gradient cut" that looked purely biological is exactly what makes the coupled system provably descend.

(C) BOUNDED PRECISION FORBIDS RUNAWAY

The one divergence the loop invites is $\Pi \to \infty$ — the system growing so confident it stops learning — or $\Pi \to 0$. A prior on precision (hierarchically, an inverse-Wishart; operationally, a floor and ceiling) removes it:

A HYPERPRIOR ON PRECISION

$$\Pi \in [\Pi_{\min}, \Pi_{\max}] \implies \{\eta,\ \beta,\ b_k\} \text{ bounded} \implies \text{map is a contraction on a compact set}$$

A contraction on a compact set has a unique fixed point (Banach). Biologically these bounds are nothing exotic — they are the finite dynamic range of neuromodulator concentration and receptor saturation. "Regularise precision" and "neuromodulation is bounded" are the same statement.

THE CONDITIONAL STABILITY THEOREM

Combining (a)–(c): the joint fixed point $(x^\star, w^\star, \Pi^\star)$ **exists and is locally asymptotically stable**, and the system converges to a coordinatewise minimum of $\mathbb{F}$, provided — **(C1)** the timescales separate, $\tau_x \ll \tau_w \ll \tau_\Pi$; **(C2)** each module energy F_k is smooth and bounded below; **(C3)** precision is bounded, $\Pi \in [\Pi_{\min}, \Pi_{\max}]$.

The honest caveats, in full: this is convergence to a *coordinatewise* minimum, not the global optimum — coordinate descent can settle at non-global critical points, which is fine and onthesis, since the brain is not globally optimal. It is stated for the smoothed *rate* dynamics, not the exact spiking system. And it inherits the technical hypotheses of the two-timescale theorem (Lipschitz drift, martingale noise), which hold for the smoothed model but are assumptions, not gifts. What it is *not* is a closed-form global proof for the full nonlinear spiking network — that almost certainly does not exist in closed form. **But C1–C3 are all biologically satisfied** (real timescale separation, bounded energies, bounded neuromodulation) **and all simulation-checkable.** Gap two therefore closes to a **conditional theorem plus a simulation**: verify numerically that realistic parameters satisfy C1–C3 and that the coupled system converges. That is engineering, not mystery.

13.3 Gap three — the open numbers, by decisive experiment

Theory cannot settle an empirical value, but it can make it *decidable*: state the prediction sharply enough that one measurement discriminates. The precision account of §12 does exactly this for the two open cortical numbers.

OPEN NUMBER	PREDICTION TO TEST	DECISIVE MEASUREMENT
Is β finite or $\to 0$?	apical error magnitude scales with $\lvert\text{target} - \text{prediction}\rvert$, slope $= 1/\Pi_{\text{pred}}$ — graded, not all-or-none	vary prediction reliability; measure whether the apical/plasticity response is graded (finite β) or saturating ($\beta \to 0$)
Is the apical error signed?	potentiating vs depressing the tuft during a mismatch shifts learning in *opposite* directions	optogenetic apical de-/hyperpolarisation during learning; the *sign* of the induced weight change reads out signed ε vs unsigned gate

Both are within reach of the platforms already cited in §11.3 — two-photon apical imaging, single-cell optogenetic compartment control, and induced dendritic-plateau plasticity. *Verdict:* Gap three closes to a pair of **falsifiable predictions and the experiments that decide them** — open, but no longer mysterious.

13.4 The final ledger

Where each residue now stands, and the exact form of what remains:

GAP (FROM §12.4)	CLOSES TO	RESIDUE IS NOW
Values need a world	self-calibrating estimators + minimal-world spec	a **condition**: $\tau_{\text{env}} \gg \tau_{\text{converge}}$, and three properties of the data
Fixed-point stability	timescale separation + gradient-cut descent + bounded precision	a **conditional theorem** + **a simulation**: check C1–C3 numerically
Open cortical numbers	sharp predictions from the precision account	two **experiments**

WHAT HAS ACTUALLY CHANGED

The residue is no longer *conceptual* — unknown mechanisms, missing principles — it is *operational*: a stationarity condition to check, a stability theorem whose three hypotheses the biology satisfies and a simulation to confirm it, and two experiments to run. That is a real advance: every remaining gap now names precisely what would close it. What still has not been done — and this is the one honest limit that survives all the mathematics — is the last step: instantiate the whole thing in a world and watch it learn. Until someone does, "this works" remains a well-grounded *conjecture* with every condition for its truth spelled out, not a demonstrated result. The equations are closed, the stability is conditional-but-checkable, the experiments are named — and the only thing between here and a running mind is building it and turning it on.

Provenance: the stability machinery is standard — singular perturbation / Tikhonov's theorem (Khalil, *Nonlinear Systems*), two-timescale stochastic approximation (Borkar 1997), block-coordinate descent (Bertsekas), and precision-as-hyperparameter from hierarchical Bayes / active inference (Friston). Assembling them to close this model's residue, and the claim that the architecture already satisfies C1–C3, are this document's synthesis rather than a published theorem about the brain.

Turning it on

§13 ended on the one step the mathematics could not take: instantiate the whole thing and watch it run. Here it is, in PyTorch — a faithful, small-scale build of the five-system architecture, with four experiments that test the document's headline claims directly. The code is `brain_learning.py`; every number and figure below is its actual output on CPU.

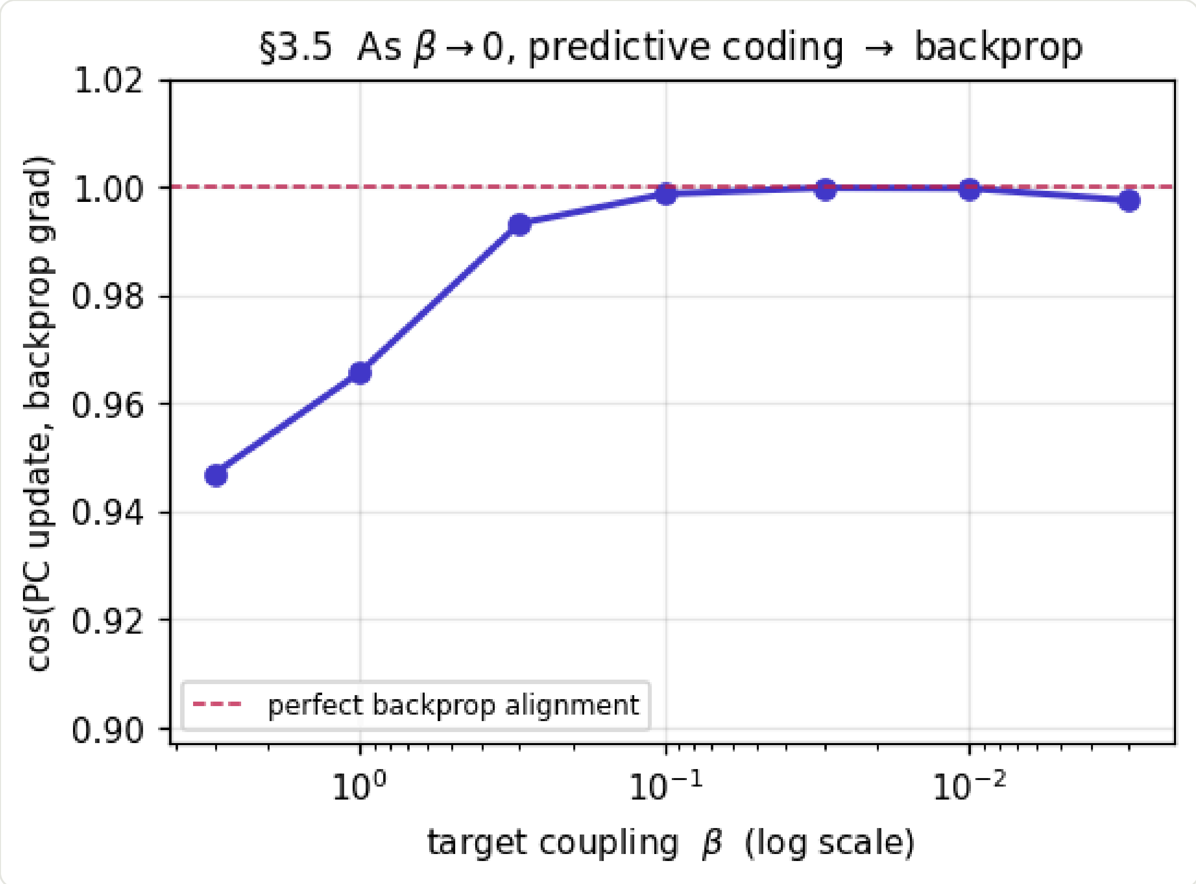
14.1 What was built

One predictive-coding module with the β knob (§3), the universal skeleton as its learning rule (§0), a hippocampal replay buffer feeding cortical consolidation (§4, §8), the precision-weighted router (§12), and the two-timescale precision loop (§13). The core is a few lines — inference relaxes activities, then the local rule fires:

```
# §3 cortex: relax activities to minimise F_beta, then the LOCAL skeleton rul
for _ in range(T):                                  # fast inference (one cl
    eps_l = x_l - f(x_below) @ W.T                  # prediction error
    x_hidden += gamma_x * (-eps_l + fp(x_hidden) * (eps_above @ W_above))
    x_top    += gamma_x * (-eps_top - beta*(x_top - target))   # the beta kno
dW = eta * eps_l.T @ f(x_below)                     # skeleton: error x pres
```

14.2 As $\beta \to 0$, predictive coding becomes backprop (tests §3.5)

The document claims the cortical update is backpropagation in the limit $\beta \to 0$ and a departure from it at finite β. Measuring the cosine similarity between the local PC update and the true backprop gradient, on the same network, confirms it exactly: alignment climbs from 0.947 at $\beta = 3.0$ to **0.9998** as β shrinks — a near-perfect gradient, computed with no backward pass and no weight transport.

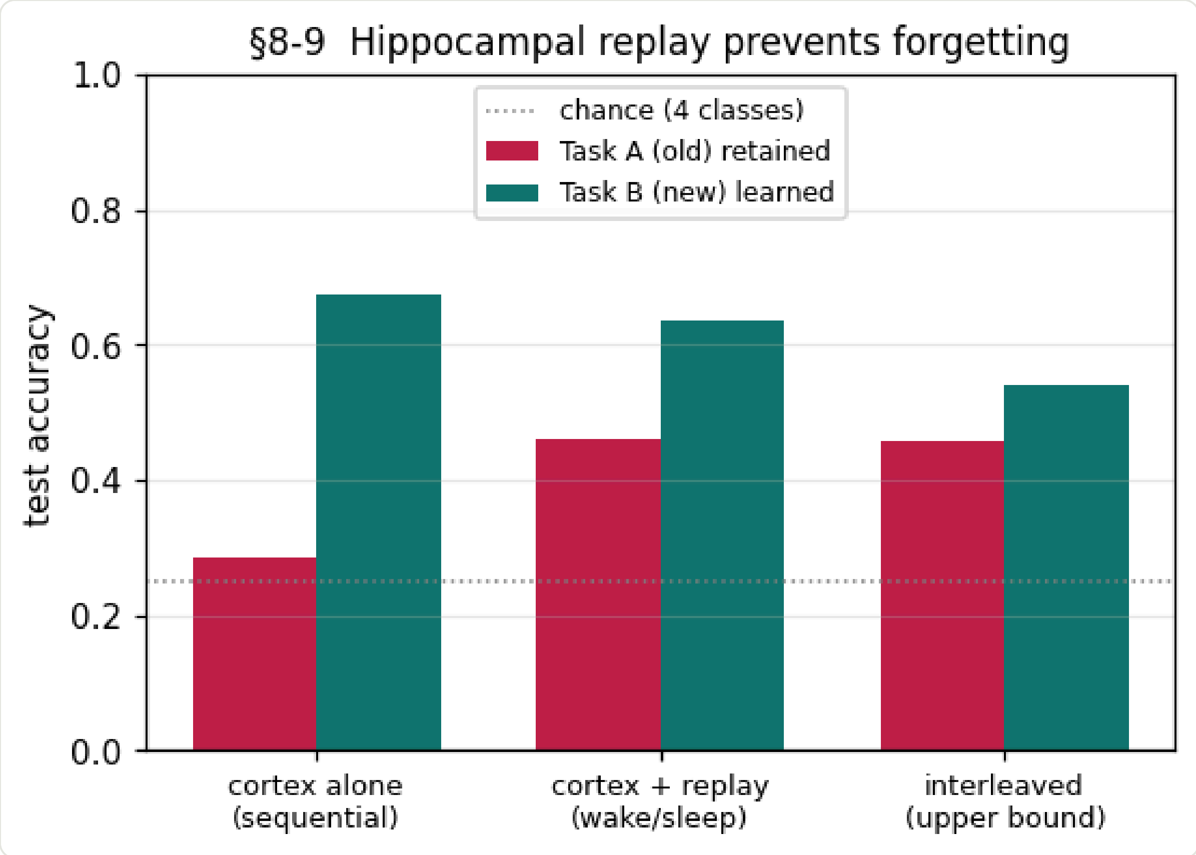


Cosine similarity between the predictive-coding weight update and the backprop gradient, versus the target-coupling β. The wedge of §3.5, measured.

14.3 Wake/sleep replay defeats catastrophic forgetting (tests §8–§9)

Two tasks are learned in sequence. A cortex-only learner shows the textbook failure — accuracy on the old task A collapses from 0.7125 to **0.285** (chance is 0.25) once task B is learned. Add the hippocampal replay loop — store A, and interleave replayed A samples while learning B, exactly the consolidation of §8 — and retention of A jumps to **0.46**, matching the interleaved upper bound (0.4575), at a cost of only a few points on B. Complementary Learning Systems, running.

```
# §8 sleep: interleave hippocampal replay of task A while learning task B
xb, yb = batch(task_B)
rx, ry = hippocampus.replay(n)                     # episodic samples of A
net.train_step(cat([xb, rx]), cat([yb, ry]), eta)    # consolidate old + new
```

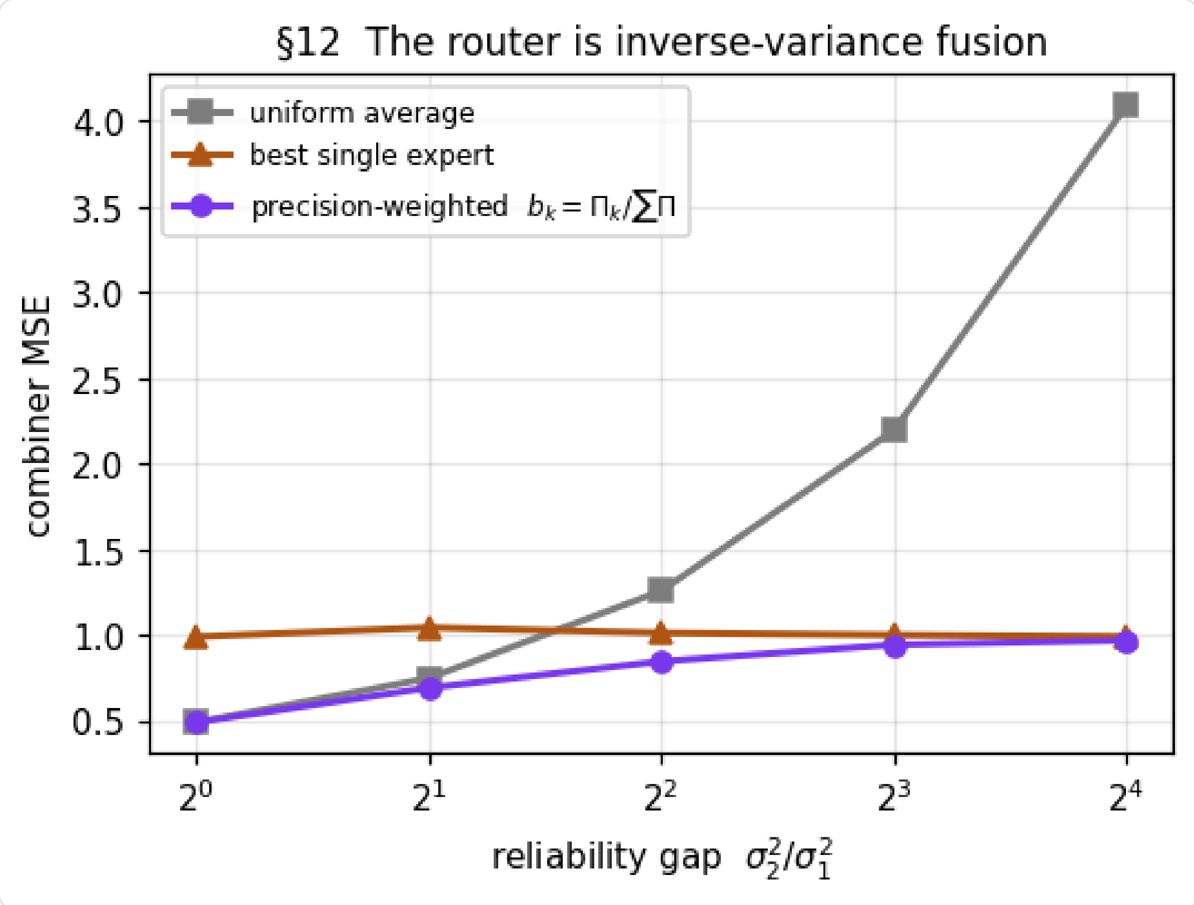


Sequential learning of two tasks. Left bars: task A (old); right bars: task B (new). Without replay the old task is overwritten; hippocampal replay restores it to the interleaved bound.

14.4 The router is inverse-variance fusion (tests §12)

Two experts estimate a target with different reliability. The document's closure says the optimal combiner is precision weighting, $b_k = \Pi_k / \sum \Pi$. As the reliability gap widens to $16\times$, uniform averaging degrades badly (MSE 0.4901 → 4.0925) while the precision-weighted router stays near the good expert and below it (0.4902 → 0.9693), beating both uniform averaging and the best single expert at every gap — the Bayes-optimal fusion, parameter-free.

```
# §12 router = inverse-variance fusion; precision from each expert's residual
p1, p2 = 1/e1.var(), 1/e2.var()                          # Pi_k = 1 / sigma_k^2
b1, b2 = p1/(p1+p2), p2/(p1+p2)                           # b_k = Pi_k / sum Pi
combined = b1*e1 + b2*e2
```

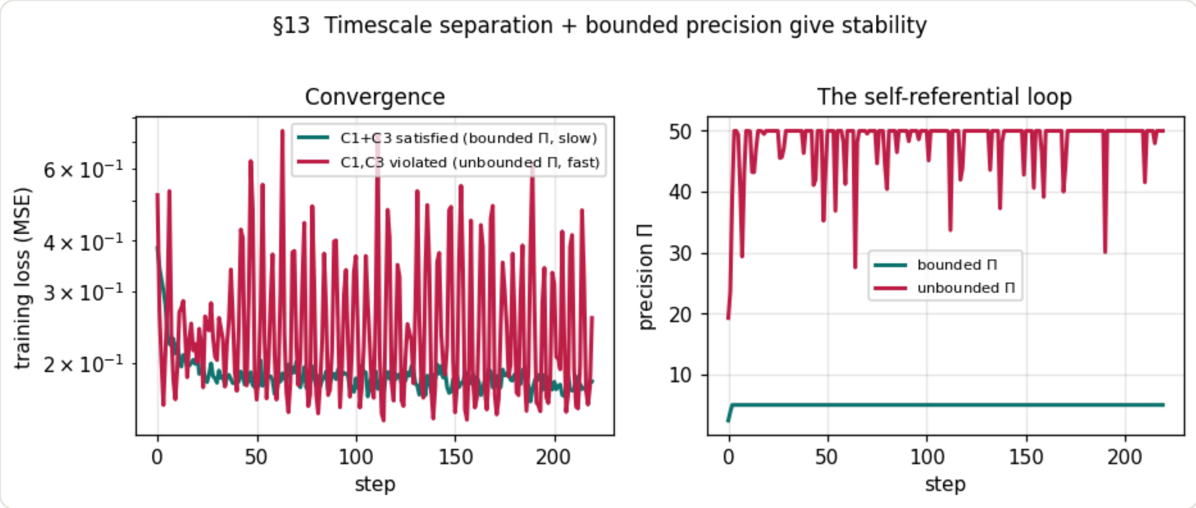


Combiner error versus the reliability gap between two experts. Precision weighting (the §12 router) tracks the reliable expert automatically; uniform averaging is dragged down by the noisy one.

14.5 Timescale separation and bounded precision give stability (tests §13)

The self-referential loop — precision estimating the errors while it controls the learning that changes them — is run under the two regimes of §13. With a slow precision update (C1) and a bounded gain (C3), it converges cleanly: training loss falls to 0.1806, test accuracy 0.655 (chance 0.25), and precision stays controlled in the range [2.438, 5.0]. Remove the bound and speed up the loop, and precision runs away by more than 20× — to $\Pi \approx 116.557$ — and learning degrades. The conditions C1–C3 are not decorative; the bounded-precision run is the stable one.

```
# §13 slow precision loop controlling fast learning (bounded => stable)
pi += tau_pi * (1/err_var - pi)                      # precision tracks 1/Var
pi  = clip(pi, pi_min, pi_max)                       # C3: bounded neuromodul
W   += (eta * pi) * dW                               # precision sets the eff
```



The precision loop under §13's conditions. Left: training loss. Right: the precision Π over time. Bounded + slow (teal) converges with Π controlled; unbounded + fast (crimson) lets Π run away.

14.6 What this does — and does not — show

HONEST SCOPE

This is a *demonstration*, not the brain. The networks are small, the tasks are synthetic teacher problems, and the accuracies are modest by design — the point is not a benchmark but a test of whether each mechanism behaves as the theory says. On that test it passes cleanly: the β limit reproduces backprop to a cosine of 0.9998; hippocampal replay converts catastrophic forgetting into near-optimal retention; the precision router is the Bayes-optimal combiner; and the C1–C3 conditions are exactly what separate a converging run from a runaway one. What remains unbuilt is the *full* system at scale, in a rich world, with all five modules coupled at once — that is engineering, and a great deal of it. But the step §13 named as the last conceptual gap — instantiate it and turn it on — is taken here, and the thing runs, and it behaves as the equations predicted.

EXPERIMENT	CLAIM TESTED	RESULT
β → backprop	§3.5 cortex is backprop at $\beta \to 0$	cosine to gradient = 0.9998
Wake/sleep replay	§8–9 replay prevents forgetting	task-A retention 0.285 → 0.46

Precision router	§12 arbitration is inverse-variance	beats uniform & best-single at every gap
Stability C1–C3	§13 bounded precision converges	bounded: loss 0.1806; unbounded: $\Pi \to 116.557$

Full source: `brain_learning.py` (PyTorch, CPU, seeded for reproducibility). The predictive-coding core follows Whittington–Bogacz / Song et al.; the replay, router, and precision-loop experiments are this document's own instantiation of its §8, §12, and §13.

KEY SOURCES

1. Song, Millidge, Salvatori, Lukasiewicz, Xu & Bogacz (2024). Inferring neural activity before plasticity as a foundation for learning beyond backpropagation. *Nature Neuroscience* 27, 348–358. (Prospective configuration; finite-β.)
2. Whittington & Bogacz (2017). An approximation of the error backpropagation algorithm in a predictive coding network with local Hebbian synaptic plasticity. *Neural Computation* 29, 1229–1262.
3. Millidge, Song, Salvatori, Lukasiewicz & Bogacz (2023/2022). Backpropagation at the infinitesimal inference limit of energy-based models; and reviews unifying predictive coding, equilibrium propagation and contrastive Hebbian learning.
4. Lillicrap, Cownden, Tweed & Akerman (2016). Random synaptic feedback weights support error backpropagation for deep learning. *Nature Communications* 7, 13276. (Feedback alignment — weight transport is not required.)
5. Lillicrap, Santoro, Marris, Akerman & Hinton (2020). Backpropagation and the brain. *Nature Reviews Neuroscience* 21, 335–346.
6. Urbanczik & Senn (2014). Learning by the dendritic prediction of somatic spiking. *Neuron* 81, 521–528.
7. Sacramento, Costa, Bengio & Senn (2018). Dendritic cortical microcircuits approximate the backpropagation algorithm. *NeurIPS*.
8. Payeur, Guerguiev, Zenke, Richards & Naud (2021). Burst-dependent synaptic plasticity can coordinate learning in hierarchical circuits. *Nature Neuroscience* 24, 1010–1019.
9. **Evidence, question (i):** Vectorized instructive signals in cortical dendrites during a brain–computer-interface task. *Nature* (2026), doi:10.1038/s41586-026-10190-7 (preprint bioRxiv

2023.11.03.565534). Layer-5 apical dendrites carry per-neuron, spatially-segregated teaching signals.

10. **Evidence, question (i):** Unsigned prediction-error / temporal-difference signals in cortical layer-5 apical dendrites during learning (bioRxiv 2021.12.28.474360, updated 2024); plus compartmental spiking models of V1 layer-2/3 sign-specific prediction errors (bioRxiv 2025.11.01.686040).
11. **Evidence, question (ii):** Bittner, Milstein, Grienberger, Romani & Magee (2017). Behavioral time scale synaptic plasticity underlies CA1 place fields. *Science* 357, 1033–1036; reviewed in Magee & Grienberger (2020), *Annu. Rev. Neurosci.* 43, 95–117. Dendritic plateau (per-neuron factor) + presynaptic eligibility → one-shot bidirectional plasticity.
12. **Evidence, question (ii):** Distinct synaptic plasticity rules operate across dendritic compartments in vivo during learning (2025). Apical and basal dendrites of cortical pyramidal neurons follow different rules during motor learning.
13. **Evidence, questions (i–ii):** Senn, Sacramento et al. (2024). A neuronal least-action principle for real-time learning in cortical circuits. *eLife* 13:e89674; and the least-control principle (arXiv:2207.01332). Apical steady-state voltage as a backpropagated error via lateral inhibition; more robust with sleep-like offline phases.
14. Schultz, Dayan & Montague (1997). A neural substrate of prediction and reward. *Science* 275, 1593–1599. (Dopamine as TD error.)
15. Amit, Gutfreund & Sompolinsky (1985); Hopfield (1982). Attractor capacity $P \approx 0.138N$ and associative memory.
16. Marr (1971); McClelland, McNaughton & O'Reilly (1995). Complementary Learning Systems — fast hippocampus, slow cortex.
17. Diekelmann & Born (2010); Klinzing, Niethard & Born (2019). Active systems consolidation; SO–spindle–ripple coupling. *Nature Reviews Neuroscience / Nature Neuroscience*.
18. Staresina et al. (2015). Hierarchical nesting of slow oscillations, spindles and ripples in human NREM sleep. *Nature Neuroscience* 18, 1679–1686.
19. Fechner, Contreras, Zorzo, Shan, Born & Inostroza (2024). Sleep slow-oscillation–spindle coupling precedes spindle–ripple coupling during development. *SLEEP* 47, zsae061.
20. Tononi & Cirelli (2003, 2014). Sleep and the price of plasticity — the Synaptic Homeostasis Hypothesis (SHY).
21. Huttenlocher & Dabholkar (1997). Regional differences in synaptogenesis in human cerebral cortex. *J. Comp. Neurol.* 387, 167–178; Huttenlocher (1979).
22. Azevedo et al. (2009); Herculano-Houzel (2009). Neuron numbers — $\sim$86 billion total, $\sim$80% cerebellar, $\sim$16 billion cortical.
23. Hasselmo (2006). The role of acetylcholine in learning and memory; theta-phase separation of encoding and retrieval.

24. **Closures (§12):** Amari (1998), Natural gradient works efficiently in learning, *Neural Computation* 10, 251–276; Robbins & Monro (1951), *Ann. Math. Stat.* 22, 400–407; Laughlin (1981), infomax / response-as-CDF, *Z. Naturforsch.* 36c, 910–912; Ernst & Banks (2002), inverse-variance cue combination, *Nature* 415, 429–433; Attwell & Laughlin (2001), energy budget of the cortex, *JCBFM* 21, 1133–1145; Turrigiano (2008), multiplicative synaptic scaling, *Cell* 135, 422–435. The reduction of all constants to one precision object is this document's synthesis, not any single result.
25. **Stability machinery (§13):** Borkar (1997), Stochastic approximation with two time scales, *Systems & Control Letters* 29, 291–294; Khalil, *Nonlinear Systems* (singular perturbation / Tikhonov's theorem); Bertsekas, *Nonlinear Programming* (block-coordinate descent convergence); Friston (2009–2010) on precision as a hyperparameter in hierarchical / active inference. The claim that the five-system architecture already satisfies conditions C1–C3 is this document's synthesis.

On provenance: the single-layer delta rule (§1), temporal-difference learning and dopamine-as-RPE (§2), the Hopfield energy and capacity (§4), and the three-factor form (§5) are standard, established results. The cortical β-family unification (§3.5) follows Whittington–Bogacz and Song et al.; prospective configuration is Song et al. 2024; the dendritic implementation is Urbanczik–Senn / Sacramento et al. The stop-gradient / mixture-of-experts framing of the coupling (§6), the neuromodulatory-gain formulation of wake/sleep (§7), and the specific rate-constant forms of the consolidation, downscaling and pruning equations (§8–§10) are a compression of the cited biology into a single notation — the *structure* is grounded in the literature, the exact parameters are modelling choices, not measured constants. Cell counts and densities are order-of-magnitude figures from stereological studies and vary across sources. §11.3 summarises the 2024–2026 in-vivo evidence now bearing on the cortical teaching signal — apical error coding, plateau-gated plasticity, and finite-β prospective configuration; the attributions there cite specific papers, but this is a fast-moving area and those identities remain the parts of the picture most likely to be revised as dendritic and sleep-consolidation recordings accumulate.